

OSA Data Assessment Report

Personal details	Gender: M	Age: 41	BMI: Not Provided
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Clinical background: Self-reported: hypertension, depression, gastroesophageal reflux (GERD), bruxism

Patient complaints: Snoring, unrefreshing sleep, teeth grinding, excessive daytime sleepiness, waking up feeling tired

Patient goals: Improve daytime energy, sleep quality, morning headaches, and reduce associated health risks

Sleep Study Data

AHI	61 (Severe OSA)
RDI	61.6
ODI	54.4
O2 Nadir	81%
Less than 90% O2	3.5%

Observations:

- Data indicates severe obstructive sleep apnea (OSA), with an AHI of 61.0 and RDI of 61.6.
- Oxygen saturation dropped to a nadir of 81%, with SpO₂ <90% during 3.5% of total sleep time.
- The Oxygen Desaturation Index (ODI) was 54.4, reflecting frequent respiratory-related desaturations.
- Findings suggest a high respiratory burden, significant sleep fragmentation, and increased cardiovascular risk.

Structural Observations from Imaging Data



Key Observations Details

Obstruction Sites Velopharyngeal, oropharyngeal, and tongue base restriction.

Soft Palate & Uvula Elongated and swollen, partially obstructing the oropharynx.

Key Observations Details

Tongue Position	Posteriorly positioned tongue base encroaching the airway; Visualization suboptimal
Bite & Jaw Structure	Edge-to-edge bite, anterior crossbite, lingually tipped mandibular and maxillary teeth, missing first premolars, and dental retrusion.
Hyoid Bone	Hyoid bone positioned significantly inferior to the mandibular plane; not visualized within the scan range.
TMJ	TMJ structures not visible in current imaging data.
Nasal & Sinus Findings	Mild to moderate mucosal thickening in the maxillary sinuses; not fully visualized within the scan range.

Note: Although the CBCT scan was suboptimal and did not fully capture all relevant anatomical regions, clinical photographs revealed an edge-to-edge bite, anterior crossbite, both anterior and posterior open bite, very narrow and underdeveloped jaw arches, and a large tongue that covers the lower posterior teeth upon opening.

Conclusion:

The provided imaging data suggests potential airway obstruction due to anatomical factors, including posterior tongue positioning, elongated soft palate and uvula, narrow dental arches with lingually tipped teeth, retruded dental position, and an inferiorly positioned hyoid bone. Additional findings from clinical photographs—such as a narrow and underdeveloped maxillomandibular arch form, edge-to-edge anterior relationship, malocclusion, and a large tongue overlapping the lower posterior teeth—further support the likelihood of compromised airway patency and should be considered in treatment planning.

Possible Treatment Considerations

Due to the combined clinical and radiographic findings, as well as the severity of the OSA, oral appliance therapy alone may not be optimal for this patient. The following treatment pathways are proposed in order of preference:

- First-line: Maxillary expansion with combined mandibular and maxillary advancement surgery.
- Second-line: CPAP therapy.
- Third-line: Combination of CPAP and oral appliance therapy.
- Fourth-line: Combination of Inspire therapy and oral appliance therapy.
- Fifth-line: Oral appliance therapy as a standalone treatment option.
 - If the patient opts for oral appliance therapy, anterior and posterior vertical opening may support airway patency by creating additional space for tongue repositioning. Encouraging nasal breathing may also be beneficial.

Device Design Data Considerations

Parameter	Data-Based Consideration
Mandibular Advancement	+2-3 mm from edge-to-edge positioning
Vertical Opening	5-6 mm anterior
Anterior Window	Large, allowing tongue access to upper incisors
Retention Features	Ball clasps for future adjustments, elastic hooks for stabilization
Material	High-density acrylic (hygienic, adjustable)
Pre-set	+1mm initial mandibular advancement
Anterior Acrylic	Full coverage
Coverage	Full coverage
Clinical Notes	Mandibular advancement should be reassessed following TMJ visualization. Ensure a minimum of 4 mm of posterior space is available when determining the final vertical opening for the oral appliance. In this case, the use of elastic bands should be avoided to allow the mandible to drop and maintain an open-mouth posture, providing additional anterior space for tongue repositioning.

Recommendations for Further Evaluation

- Sleep specialist consultation is recommended to determine the most appropriate long-term treatment strategy.
- ENT evaluation is advised to assess the maxillary sinus findings noted on imaging.
- The available sleep study data is incomplete; retrieving the full report is recommended, including positional data, to evaluate the potential benefit of positional therapy.
- BMI, height, and weight data are missing; this information should be obtained to assess the need for a weight loss program as part of the treatment plan.
- A complementary CBCT scan is recommended to assess the temporomandibular joints, nasal cavity, and maxillary sinuses. This may inform ENT intervention, use of additional adjunctive tools such as nasal cones, and final mandibular advancement based on condylar position.
- If oral appliance therapy is selected as a temporary or combined treatment approach, it is important to align expectations with the patient and clarify that oral appliance therapy alone is unlikely to be sufficient. Successful treatment will require patient commitment to adjunctive interventions such as weight loss, sleep hygiene, and positional therapy.

- A new intraoral scan is recommended once final mandibular advancement and vertical opening parameters are established.

Oral Appliance Options for Consideration

Device		Key Features
Emerald Herbst		Strong, lightweight, gold binding embedded, hard high-density acrylic for hygiene and durability
Respire Herbst Pink AT		Promotes mandible stabilization, high-density acrylic, metal mesh embedded
Daynaflex Herbst		Accurate fit, enhanced tongue space, stain-resistant PMMA

Disclaimer

This AI-generated report is intended as a data assessment tool to support clinical decision-making. It does not constitute a medical diagnosis, treatment recommendation, or radiological interpretation. All imaging findings, including CBCT references, should be reviewed and confirmed by a certified radiologist or qualified healthcare provider. Final treatment decisions must be made by licensed professionals based on a comprehensive clinical evaluation. No liability is assumed for medical decisions made solely on the basis of this report.